

13 (a) On the grid show, by shading, the region that satisfies all these inequalities.

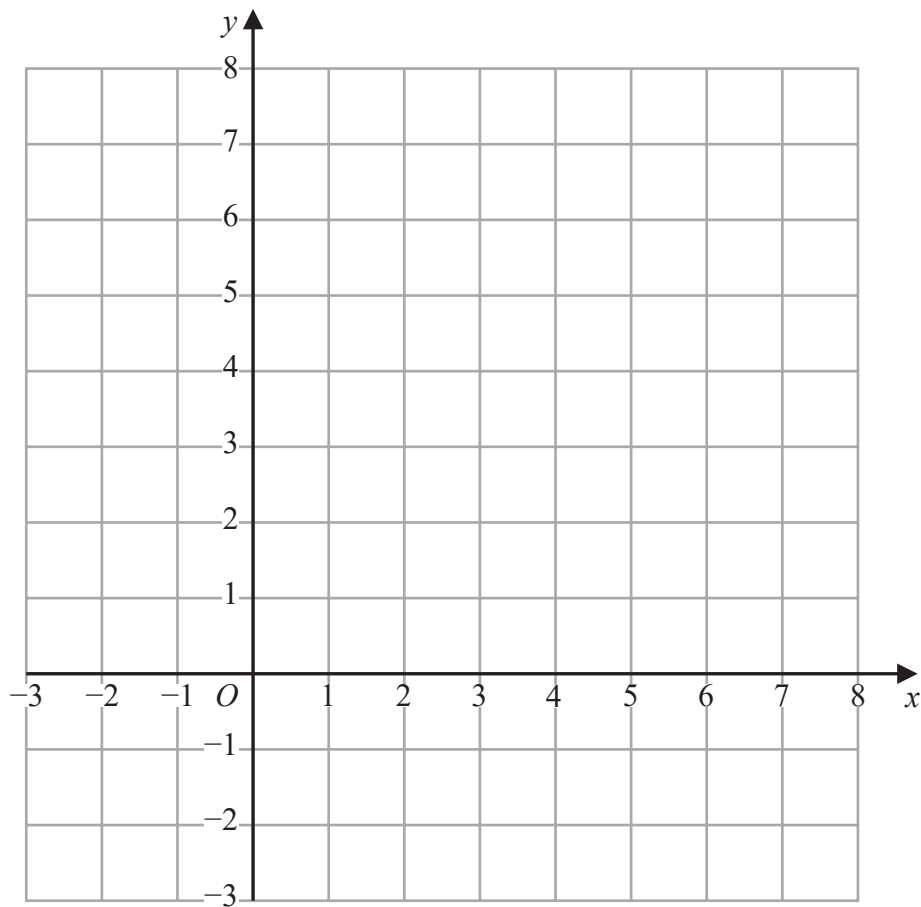
$$x \geq 0$$

$$x \leq 2$$

$$y \leq x + 3$$

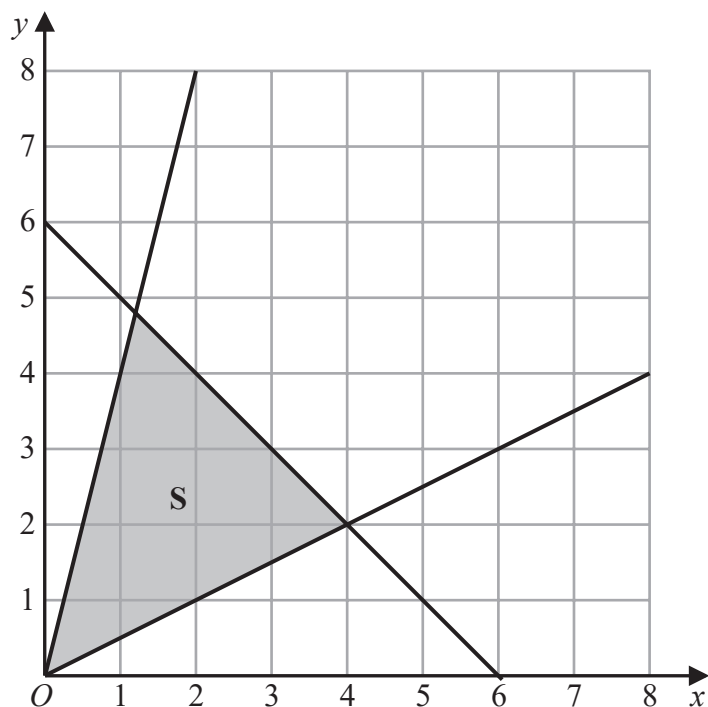
$$2x + 3y \geq 6$$

Label the region **R**.



(b) The diagram below shows the region **S** that satisfies the inequalities

$$y \leq 4x \quad y \geq \frac{1}{2}x \quad x + y \leq 6$$



Geoffrey says that the point with coordinates (2, 4) does not satisfy all the inequalities because it does not lie in the shaded region.

Is Geoffrey correct?

You must give a reason for your answer.

(1)

(Total for Question 13 is 5 marks)